

LAST NAME: _____

Practice Quiz 05A — MATH 2055 — Fall 2026

Practice for Quiz 05

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

Write in full sentences. Introduce the arbitrary element and name your proof strategy.

1. Prove

$$\forall x \in \mathbf{Z} \ (4 \nmid x^2 \rightarrow 2 \nmid x).$$